MSG No. 124.359 $P4/m'c'c'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `422`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,6,9,10,11,14]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `422`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `4/m'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,2,3,6,9,10,11,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,9,14]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,10,11]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,5,12,13]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,8,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,7,8]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,12,13,14]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[10,11,15,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,7,8]`
3	$[0,\ 0,\ -z]$	`[9,10,11,14]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[12,13,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,3,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,10,11,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13,15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[10,11]`
7	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13]`
8	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[15,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,5]`
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,4]`
4	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[6,8]`
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[9,15]`
6	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[10,13]`
7	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[11,12]`
8	$\left[x,\ x,\ \frac{3}{4}\right]$	`[14,16]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,4]`
2	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,7]`

continued ...

Table 11

No.	position	mapping
3	$\left[0,\ -x,\ \frac{1}{4}\right]$	[3,8]
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[5,6]
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[9,12]
6	$\left[0,\ -x,\ \frac{3}{4}\right]$	[10,15]
7	$\left[0,\ x,\ \frac{3}{4}\right]$	[11,16]
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	[13,14]

Table 12: Wyckoff site: 8l, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,4]
2	$\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[2,7]
3	$\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,8]
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[5,6]
5	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[9,12]
6	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[10,15]
7	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[11,16]
8	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[13,14]

Table 13: Wyckoff site: 8m, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-y,\ x,\ 0]$	[2,11]
3	$[y,\ -x,\ 0]$	[3,10]
4	$\left[x,\ -y,\ \frac{1}{2}\right]$	[4,13]
5	$\left[-x,\ y,\ \frac{1}{2}\right]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]
7	$\left[y,\ x,\ \frac{1}{2}\right]$	[7,16]
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[8,15]

Table 14: Wyckoff site: 16n, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[4]
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]

continued ...

Table 14

No.	position	mapping
10	$[y,\, -x,\, -z]$	[10]
11	$[-y,\, x,\, -z]$	[11]
12	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]